Chemistry Section A

Section Id :	691121101
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	691121101
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Id : 6911211251 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following contain the same number of atoms ?

(Given : Molar mass in g mol^{-1} of H, He, O and S are 1, 4, 16 and 32 respectively)

A. 2 g of O_2 gas

B. 4 g of SO_2 gas

C. 1400 mL of O_2 at STP

D. 0.05 L of He at STP

E. 0.0625 mol of H_2 gas

Choose the **correct** answer from the options given below :

Options :

6911214251. A and B only

6911214252. B and C only

6911214253. C and D only

6911214254. A, C and E only

Question Number : 52 Question Id : 6911211252 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The Bohr radius of a hydrogen like species is 70.53 pm. The species and the stationary state (n) are respectively

(Given : Hydrogen atom Bohr radius is 52.9 pm)

Options :

6911214255. Li^{2+}, 3

6911214256. He^{+}, 3

6911214257. He^+, 2

6911214258. Li^{2+}, 2

Question Number : 53 Question Id : 6911211253 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : The number of compounds among SO_2, SO_3, SF_4, SF_6 and H_2S in which sulphur does not obey the Octet rule is 3.

Statement II : Among [H_2O, ClF_3, SF_4], [NH_3, BrF_5, SF_4], [BrF_5, ClF_3, XeF_4] and [XeF_4, ClF_3, H_2O], the number of sets in which all the molecules have one lone pair of electrons on the central atom is 1.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911214259. Both **Statement I** and **Statement II** are true

6911214260. Both **Statement I** and **Statement II** are false

6911214261. **Statement I** is true but **Statement II** is false

6911214262. **Statement I** is false but **Statement II** is true

Question Number : 54 Question Id : 6911211254 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match **List - I** with **List - II**.

Given V_1 and V_2 are initial and final volumes respectively.

	List - I (Isothermal process)		**List - II (Expression)**
A.	Reversible expansion	I.	$q = 0$
B.	Free expansion	II.	$q = nRT \ln \frac{V_2}{V_1}$
C.	Irreversible Compression	III.	$w = -p_{ext}(V_1 - V_2)$
D.	Cyclic reversible	IV.	$\frac{q_{rev}}{T} = 0$

Choose the **correct** answer from the options given below :

Options :

6911214263. A-II, B-III, C-I, D-IV

6911214264. A-II, B-I, C-IV, D-III

6911214265. A-II, B-I, C-III, D-IV

6911214266. A-I, B-II, C-III, D-IV

Question Number : 55 Question Id : 6911211255 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Chamber 1	Semi-permeable membrane	Chamber 2
18 g glucose in 100 mL aqueous solution		30 g glucose in 250 mL aqueous solution

Statement I : H_2O molecules move from the chamber 1 to chamber 2.

Statement II : The osmotic pressure of a solution prepared by dissolving 50 mg of potassium sulphate (molar mass = 174 g/mol) in 2 L of water (at 27 °C) is 0.0107 bar. (Given: $R = 0.083\ dm^3\ bar\ K^{-1}\ mol^{-1}$ and assume complete dissociation of electrolyte)

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911214267. Both **Statement I** and **Statement II** are true

6911214268. Both **Statement I** and **Statement II** are false

6911214269. **Statement I** is true but **Statement II** is false

6911214270. **Statement I** is false but **Statement II** is true

Question Number : 56 Question Id : 6911211256 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given is a concentrated solution of a weak electrolyte A_xB_y of concentration 'c' and dissociation constant 'K'. The degree of dissociation is given by :

Options :

6911214271. $\left[K \times c^{x+y-1} x^x y^y\right]^{x+y}$

6911214272. $\left(\frac{K}{c^{x+y-1}\ x^x y^y}\right)^{\frac{1}{x+y}}$

6911214273. $\left(\frac{c^{x+y-1}\ x^x y^y}{K}\right)^{x+y}$

6911214274. $\left(\frac{c^{x+y-1}\,x^x y^y}{K}\right)^{\frac{1}{x+y}}$

Question Number : 57 Question Id : 6911211257 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For a general redox reaction

Anode $\quad Red_1 \rightarrow Ox_1^{n_1^+} + n_1 e^-$

Cathode $\quad Ox_2 + n_2 e^- \rightarrow Red_2^{n_2^-}$

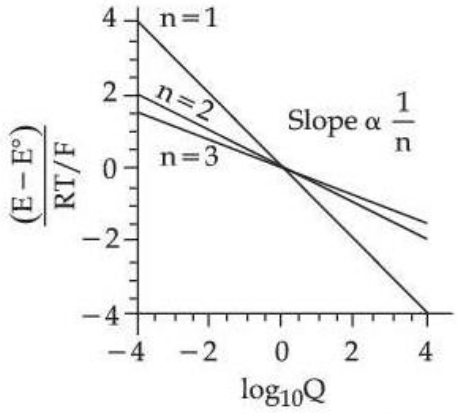
Which of the following statement is **incorrect** ?

Options :

6911214275. The overall reaction can be written as $n_2 Red_1 + n_1\ Ox_2 \rightleftharpoons n_2 Ox_1^{n_1^+} + n_1 Red_2^{n_2^-}$

6911214276.

The electrons do not appear in the overall reaction because electrons produced at the anode are consumed at the cathode.



6911214277. Here n is the number of electrons transferred in redox reaction.

6911214278.

If the reaction is carried out reversibly, the electrical work done is equal to the ratio of charge and potential difference through which charge is moved.

Question Number : 58 Question Id : 6911211258 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In a period, the first ionisation enthalpy of the element at extreme left and the negative electron gain enthalpy of the extreme right element, except noble gases, are respectively.

Options :

6911214279. lowest and lowest

6911214280. highest and lowest

6911214281. lowest and highest

6911214282. highest and highest

Question Number : 59 Question Id : 6911211259 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : $F_2O < H_2O < Cl_2O$ is the correct trend in terms of bond angle.

Statement II : SiF_4, SnF_4 and PbF_4 are ionic in nature.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911214283. Both **Statement I** and **Statement II** are true

6911214284. Both **Statement I** and **Statement II** are false

6911214285. **Statement I** is true but **Statement II** is false

6911214286. **Statement I** is false but **Statement II** is true

Question Number : 60 Question Id : 6911211260 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The correct order of first (Δ_iH_1) and second (Δ_iH_2) ionisation enthalpy values of Cr and Mn are :

A. Δ_iH_1 : Cr > Mn

B. Δ_iH_2 : Cr > Mn

C. Δ_iH_1 : Mn > Cr

D. Δ_iH_2 : Mn > Cr

Choose the **correct** answer from the options given below :

Options :

6911214287. A and B only

6911214288. B and C only

6911214289. A and D only

6911214290. C and D only

Question Number : 61 Question Id : 6911211261 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following sequences of hybridisation, geometry and magnetic nature are **correct** for the given coordination compounds ?

A. $[NiCl_4]^{2-}$ – sp^3, tetrahedral, paramagnetic

B. $[Ni(NH_3)_6]^{2+}$ – sp^3d^2, octahedral, paramagnetic

C. $[Ni(CO)_4]$ – sp^3, tetrahedral, paramagnetic

D. $[Ni(CN)_4]^{2-}$ – dsp^2, square planar, diamagnetic

Choose the **correct** answer from the options given below :

Options :

6911214291. A, B, C and D

6911214292. B, C and D only

6911214293. A, C and D only

6911214294. A, B and D only

Question Number : 62 Question Id : 6911211262 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : A mixture of $C_{12}H_{22}O_{11}$ (sugar) and NaCl can be separated by dissolving sugar in alcohol, due to differential solubility.

Statement II : Rose essence from rose petals is seperated by steam distillation due to its high volatility and insolubility in H_2O.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911214295. Both **Statement I** and **Statement II** are true

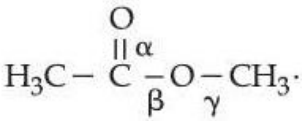
6911214296. Both **Statement I** and **Statement II** are false

6911214297. **Statement I** is true but **Statement II** is false

6911214298. **Statement I** is false but **Statement II** is true

Question Number : 63 Question Id : 6911211263 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Shown below is the structure of methyl acetate with three different α, β and γ carbon - oxygen bonds.



The correct order of bond lengths of these bonds is :

Options :

6911214299. $\alpha > \beta > \gamma$

6911214300. $\alpha < \beta < \gamma$

6911214301. $\alpha = \beta = \gamma$

6911214302. $\alpha < \beta = \gamma$

Question Number : 64 Question Id : 6911211264 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

'*x*' is the product which is obtained by the hydrolysis of prop-1-yne in the presence of mercuric sulphate under dilute acidic medium at 333 K. '*y*' is the product which is obtained by the reaction of ethane nitrile with methyl magnesium bromide in dry ether followed by hydrolysis. IUPAC name of product obtained from '*x*' and '*y*' in the presence of barium hydroxide followed by heating is :

Options :

6911214303. 2 - Methylpent-4-en-3-one

6911214304. 4 - Methylpent-3-en-2-one

6911214305. 4 - Methylpent-1-ene

6911214306. 2 - Methylpent-3-one

Question Number : 65 Question Id : 6911211265 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An optically active alkyl bromide C_4H_9Br, reacts with ethanolic KOH to form major compound [A] which reacts with bromine to give compound [B]. Compound [B] reacts with ethanolic KOH and sodamide to give compound [C]. One molecule of water adds to compound [C] on warming with mercuric sulphate and dilute sulphuric acid at 333 K to form compound [D]. The functional group in compound D will be confirmed by :

Options :

6911214307. Haloform test

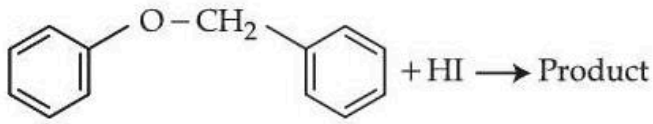
6911214308. Lucas test

6911214309. Silver mirror test

6911214310. Benedict test

Question Number : 66 Question Id : 6911211266 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following reaction.



Statement I : In the above reaction, product formed will be a mixture of benzyl alcohol and iodobenzene.

Statement II : In the above reaction, the $-O-CH_2-$ bond is cleaved to give the product.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911214311. Both **Statement I** and **Statement II** are true

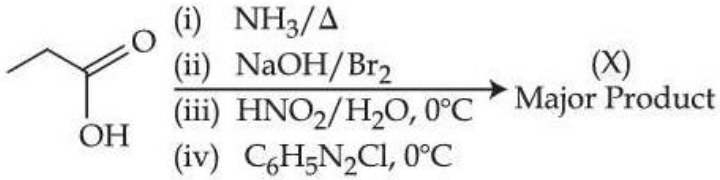
6911214312. Both **Statement I** and **Statement II** are false

6911214313. **Statement I** is true but **Statement II** is false

6911214314. **Statement I** is false but **Statement II** is true

Question Number : 67 Question Id : 6911211267 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following organic reaction sequence. Choose the final product (X) from the following (consider the major product in all intermediate reactions)



Options :

6911214315. Benzene

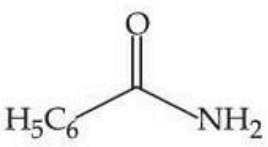
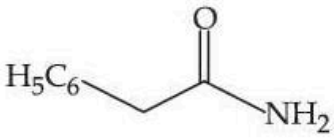
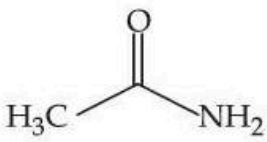
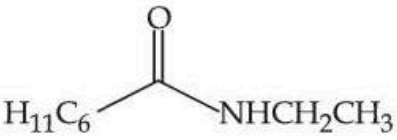
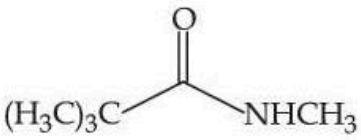
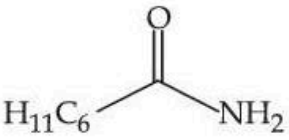
6911214316. Phenol

6911214317. Propanol

6911214318. Chlorobenzene

Question Number : 68 Question Id : 6911211268 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of compounds from the following which can undergo reaction with Br_2/KOH (alcoholic) to give respective products and these respective products can also be obtained separately by Gabriel phthalimide reaction is :

Options :

6911214319. 5

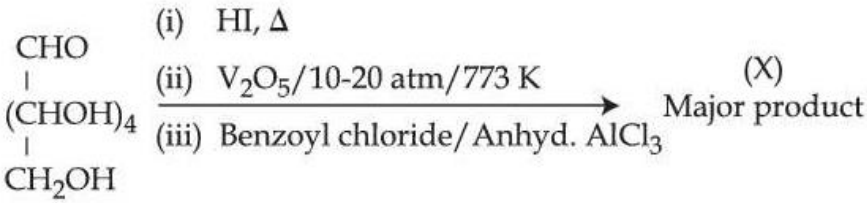
6911214320. 4

6911214321. 3

6911214322. 6

Question Number : 69 Question Id : 6911211269 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following reactions. Total number of electrons in the π bonds and lone pair of electrons in the product (X) is :



Options :

6911214323. 12

6911214324. 16

6911214325. 14

6911214326. 18

Question Number : 70 Question Id : 6911211270 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Treatment of a gas 'X' with a freshly prepared ferrous sulphate solution gives a compound 'Y' as a brown ring. The compounds X and Y are.

Options :

6911214327. NO and $[Fe(NO)]SO_4$

6911214328. NO_2 and $[Fe(NO_2)]SO_4$

6911214329. N_2O and $[Fe(N_2O)]SO_4$

6911214330. N_2O_4 and $[Fe(N_2O_4)]SO_4$

Chemistry Section B

Section Id :	691121102
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	691121102
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 71 Question Id : 6911211271 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

An excess of $AgNO_3$ is added to 100 mL of a 0.05 M solution of tetraaquadichloridochromium (III) chloride. The number of moles of AgCl precipitated will be _____ $\times 10^{-3}$.

(Nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 72 Question Id : 6911211272 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

An alkane (Y) requires 8 moles of oxygen for complete combustion and on chlorination with $Cl_2/h\nu$, (Y) gives only one monochlorinated product (Z). The total number of primary carbon atoms in (Y) is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 73 Question Id : 6911211273 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

500 mL of 0.2 M MnO_4^- solution in basic medium when mixed with 500 mL of 1.5 M KI solution, oxidises iodide ions to liberate molecular iodine. This liberated iodine is then titrated with a standard x M thiosulphate solution in presence of starch till the end point. If 300 mL of thiosulphate was consumed, then the value of x is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 74 Question Id : 6911211274 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

In a closed flask at 600 K, one mole of $X_2Y_4(g)$ attains equilibrium as given below :

$$X_2Y_4(g) \rightleftharpoons 2XY_2(g)$$

At equilibrium, 75% $X_2Y_4(g)$ was dissociated and the total pressure is 1 atm. The magnitude of $\Delta_r G^{\ominus}$ (in kJ mol^{-1}) at this temperature is _____. (Nearest Integer)
(Given : $R = 8.3$ J mol^{-1} K^{-1}; $\ln 10 = 2.3$, $\log 2 = 0.3$, $\log 3 = 0.48$, $\log 5 = 0.69$, $\log 7 = 0.84$)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 75 Question Id : 6911211275 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Decomposition of a hydrocarbon follows the equation $k = (5.5 \times 10^{11} s^{-1})\ e^{\frac{-28000K}{T}}$. The activation energy of reaction is _____ kJ mol^{-1}. (Nearest Integer)
Given : $R = 8.3$ J K^{-1} mol^{-1}

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1